WPR - 8

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY
B.Tech (CSE - AIML) VII Semester

Project Area -- Generative AI & AI Safety (Trustworthy AI / MLOps)

Students Weekly Progress Report (WPR — Week 8) For Odd Semester of session 2026-2027

To be filled by Students	Group No. 50
Students Name	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh
Roll no.	1. 569 2. 166 3. 191
Enrollment no.	1. A2305223569 2. A2305223166 3. A2305223191
Project Title finalized, if Yes, give name, if NO, give reason	Yes. Title: "GuardShield AI: Real-Time Sidecar Proxy for LLM Hallucination Detection & Prompt Injection Defense"
Synopsis submitted	Yes (Approved for Group No. 50)
Literature review	Applied to comparative empirical benchmarking against baseline guardrails (Meta Llama-Guard-3 and SelfCheckGPT). Analyzed latency, throughput, and compute constraints.
Technical & Economical Feasibility	Feasible. Python 3.11+, PyTorch CUDA (RTX 3050), FastAPI async proxy, DeBERTa-v3 SLM (86M). Zero monetary cost
Bill of Material	NA
Project Progress Schedule (PERT Chart)	• W1-W5: Setup, dataset curation, DeBERTa fine-tuning & evaluation (Completed) • W6-W8: Streaming entropy engine, NLI grounding & comparative benchmarking (Completed / Week 8) • W9-W12: Multi-turn stress testing, IEEE research paper compilation & final project viva preparation
Design of critical components	1. Comparative Benchmarking Engine (`benchmark_comparative.py`) 2. Baseline Comparison Matrix against Meta Llama-Guard-3 & SelfCheckGPT 3. Latency, VRAM footprint, and cost-per-query comparative evaluation table
Fabrication work (give %)	0% (Software Project)
Experimental work (give %)	85% (Comparative empirical evaluation completed; established 98% latency reduction over 8B guardrails while retaining 86.96% precision on adversarial injections)
Result and Analysis	Demonstrated GuardShield proxy achieving ~29-48ms latency overhead compared to 1650ms for Llama-Guard-3 and 3400ms for SelfCheckGPT; VRAM utilization reduced from 16GB to <500MB.

Report writing	WPR Week 8 drafted.
Signature of students	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh

Work done in this week (Week 8):

1. Comparative Guardrail Benchmarking: Developed `src/benchmark_comparative.py` to empirically benchmark GuardShield AI against leading industry baselines (Meta Llama-Guard-3, SelfCheckGPT, and Dense Continuous Cross-Encoders).
2. Latency & Resource Profiling: Validated that GuardShield achieves a 98.2% latency reduction over 8B guardrails (48ms vs. 1,650ms) while fitting within < 500 MB VRAM (compatible with edge/laptop hardware at ₹0 cost).
3. Streaming Support Advantage: Demonstrated that token-level Shannon entropy enables early mid-sentence hallucination termination, overcoming the 3,400ms multi-sample sampling penalty of offline detectors like SelfCheckGPT.
4. Milestone Verification: Successfully consolidated all experimental results into `data/processed/comparative_benchmark_results.json` in preparation for the IEEE research paper methodology and results sections.

To be filled by Guide (strike off whichever is not applicable)
Performance of students is satisfactory
Performance of students is unsatisfactory
A warning to be issued to student(s) (Name)
Student was not well (Name)
Date 16/09/2026 Signature of Guide